

EM HOMEWORK 4

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3:38 PM

SECTION 1.3:

4 $y' \sin 2\pi x = \pi y \cos 2\pi x$

$$\frac{dy}{dx} \sin(2\pi x) = \pi y \cos(2\pi x)$$

$$\frac{1}{y} \cdot dy = \frac{\pi \cos(2\pi x)}{\sin(2\pi x)} \cdot dx$$

$$\int \frac{1}{y} \cdot dy = \int \pi \cot(2\pi x) \cdot dx$$

$$\ln(y) = \cancel{\pi} \cdot \frac{1}{2\cancel{\pi}} \ln|\sin(2\pi x)| + C$$

$$y = C e^{\ln|\sin(2\pi x)|^{1/2}} = C (\sin(2\pi x))^{1/2}$$

10 $xy' = x + y$ (Set $y/x = u$)

$$xy' = x + y \quad \text{--- (A)}$$

let, $y = ux$

then $\frac{dy}{dx} = xu' + u$

$$\frac{dy}{dx} = x \cdot \frac{du}{dx} + u$$

$$\frac{dy}{dx} = x \cdot \frac{du}{dx} + u$$

Substitute in (A)

$$x \left(x \frac{du}{dx} + u \right) = x + ux$$

$$du = \frac{1}{x} \cdot dx$$

integrate :-

$$u = \ln|x| + C$$

$$\frac{y}{x} = \ln|x| + C$$

$$\therefore u = y/x$$

$$y = x \ln|x| + C$$

Solve the IVP. Show the steps of derivation, beginning with the general solution.

1. $xy' + y = 0, \quad y(4) = 6$

$$x \cdot \frac{dy}{dx} = -y$$

$$\frac{-1}{y} \cdot dy = \frac{1}{x} \cdot dx$$

$$\frac{1}{y} \cdot dy = -\frac{1}{x} \cdot dx$$

Integrate on both sides:-

$$\ln|y| = -\ln|x| + \ln|C|$$

$$\ln|y| = \ln\left|\frac{C}{x}\right|$$

$$\ln|y| = \ln\left|\frac{C}{x}\right|$$

$$y = C/x$$

$$y(4) = 6 \quad 6 = \frac{C}{4}$$

$$\boxed{C = 24}$$

so,

$$\boxed{y = \frac{24}{x}}$$

17. $xy' = y + 3x^4 \cos^2(y/x), \quad y(1) = 0$
(Set $y/x = u$)

$$\frac{dy}{dx} = xu' + u$$

$$x(xu' + u) = y + 3x^4 \cos^2(u)$$

$$x(xu' + u) = ux + 3x^4 \cos^2(u)$$

$$u' = 3x^3 \cos^2(u)$$

$$\int \frac{1}{\cos^2 u} \cdot du = \int 3x^3 \cdot dx$$

$$\tan u = x^3 + C$$

$$u = \tan^{-1}(x^3 + C)$$

$$\frac{y}{x} = \tan^{-1}(x^3 + C)$$

$$y = x \cdot \tan^{-1}(x^3 + C)$$

$$0 = 1 + C$$

$$\boxed{C = -1}$$

So,

$$\boxed{\therefore y = x \cdot \tan^{-1}(x^3 - 1)}$$

21. Radiocarbon dating. What should be the $^{14}_6\text{C}$ content (in percent of y_0) of a fossilized tree that is claimed to be 3000 years old? (See Example 4.)

$$\frac{dy}{dt} = ky$$

So, $y(t) = y_0 e^{kt}$

$$y(h) = y_0 e^{kh}$$

$$\frac{y_0}{2} = y_0 e^{kh}$$

$$e^{kh} = \frac{1}{2}$$

$$k = \frac{\ln(1/2)}{h}$$

• at $t = 3000$,

$$y(3000) = y_0 e^{\frac{\ln(1/2)}{h} (3000)}$$

$$= y_0 \left(\frac{1}{2}\right)^{3000/h}$$

$$h = 5715.1$$

$$\frac{y(3000)}{y_0} = \left(\frac{1}{2}\right)^{\frac{3000}{5715.1}}$$

or 69.5% of y_0

so, the fossil has 69.5% of its original carbon content.

16. Gompertz growth in tumors. The Gompertz model is $y' = -Ay \ln y$ ($A > 0$), where $y(t)$ is the mass of tumor cells at time t . The model agrees well with clinical observations. The declining growth rate with increasing $y > 1$ corresponds to the fact that cells in the interior of a tumor may die because of insufficient oxygen and nutrients. Use the ODE to discuss the growth and decline of solutions (tumors) and to find constant solutions. Then solve the ODE.

for constant solution we know that $y' = 0$

$$0 = -A y \ln(y)$$

$$\begin{array}{ll} -Ay = 0 & \ln y = 0 \\ y = 0 & y = 1 \end{array}$$

$\ln y < 0$ } $-Ay \ln y$ will be +ve, solution increases.

for $y > 1$, $\ln y > 0$ } $-Ay \ln y$ will be -ve, solution decreases.

solve,

$$\frac{dy}{dt} = -Ay \ln y$$

$$\int \frac{1}{y \cdot \ln y} \cdot dy = \int -A \cdot dt$$

$$\ln |\ln y| = -At + C$$

$$\ln |y| = C_1 e^{-At}$$

$$y = e^{C_1 e^{-At}}$$

for y_0 :-

$$y(0) = e^{C_1}$$

$$C_1 = \ln |y_0|$$

So,

$$y(t) = e^{\ln(y) \cdot e^{-At}}$$

SECTION 1.4

$$1. \quad 2xy \, dx + x^2 \, dy = 0$$

$$M = 2xy \quad N = x^2$$

$$\frac{\partial M}{\partial y} = 2x \quad \frac{\partial N}{\partial x} = 2x$$

its exact.

$$\begin{aligned} F(x, y) &= \int M \, dx \\ &= \int 2xy \, dx \\ &= y \cdot x^2 + g(y) \end{aligned}$$

$$F(x, y) = x^2 y + g(y) \quad \text{--- (1)}$$

• Taking derivative wrt y :-

$$F_y = x^2 + g'(y)$$

$$x^2 = x^2 + g'(y)$$

$$0 = g'(y)$$

So is,

So is,

$$g(y) = 0$$

so. ① becomes:-

$$F(x, y) = x^2 y$$

and general:-

$$C = x^2 y$$

$$\text{2. } x^3 dx + y^3 dy = 0$$

$$M = x^3$$

$$N = y^3$$

$$\frac{\partial M}{\partial y} = 0$$

$$\frac{\partial N}{\partial x} = 0$$

is exact!

$$F(x, y) = \int x^3 \cdot dx$$

$$F(x, y) = \frac{x^4}{4} + g(y) \quad \text{--- ①}$$

derivative wrt y:-

$$F_y = 0 + g'(y)$$

$$y^3 = g'(y)$$

Taking $\int$ wrt y:-

$$\frac{dy}{y} = g(y)$$

$$F(x, y) = \frac{x^4}{4} + \frac{y^4}{4}$$

$$C = \frac{x^4 + y^4}{4}$$

$$\boxed{C = x^4 + y^4}$$

3. $\sin x \cos y \, dx + \cos x \sin y \, dy = 0$

$$M = \sin x \cos y \quad N = \cos x \sin y$$

$$\frac{\partial M}{\partial y} = -\sin x \cdot \sin y \quad \frac{\partial N}{\partial x} = -\sin x \cdot \sin y$$

exact!

$$F(x, y) = \int \sin x \cos y \cdot dx$$

$$F(x, y) = -\cos x \cos y + g(y) \quad \text{--- (1)}$$

Taking derivative wrt y :-

$$F_y = \cos x \sin y + g'(y)$$

$$\cos x \sin y = \cos x \sin y + g'(y)$$

$$g'(y) = 0$$

$$g(y) = 0$$

eq ① becomes:-

$$F(x, y) = -\cos x \cos y$$

$$C = -\cos x \cos y$$

$$C = \cos x \cos y$$

4. $e^{3\theta}(dr + 3r d\theta) = 0$

$$M = e^{3\theta}$$

$$N = 3r e^{3\theta}$$

$$\frac{\partial M}{\partial \theta} = 3e^{3\theta}$$

$$\frac{\partial N}{\partial r} = 3e^{3\theta}$$

exact

$$F(r, \theta) = \int e^{3\theta} \cdot dr$$

$$F(r, \theta) = r e^{3\theta} + g(\theta) \text{ --- ①}$$

Taking derivative wrt θ .

$$F_a = 3re^{3a} + g'(a)$$

$$\cancel{3re^{3a}} = \cancel{3re^{3a}} + g'(a)$$

$$g'(a) = 0$$

$$g(a) = 0$$

eq ①

$$F(r, a) = re^{3a}$$

$$C = re^{3a}$$

P. $(x^2 + y^2)dx - 2xy dy = 0$

$$M = x^2 + y^2$$

$$N = -2xy$$

$$\frac{\partial M}{\partial y} = 2y$$

$$\frac{\partial N}{\partial x} = -2y$$

Non-exact

$$u(x) = e. \int \frac{\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}}{N} \cdot dx$$

$$= e \int \frac{2y - (-2y)}{-2xy} \cdot dx$$

$$= \cancel{e} (\ln |x|^{-2})$$

$$= \frac{1}{x^2}$$

So,

$$M = \frac{x^2 + y^2}{x^2}$$

$$N = \frac{-2xy}{x^2}$$

$$= 1 + \frac{y^2}{x^2}$$

$$= \frac{-2y}{x}$$

$$\frac{\partial M}{\partial y} = \frac{2y}{x^2}$$

$$\frac{\partial N}{\partial x} = (-2y)(-1)(x^{-2}) = \frac{2y}{x^2}$$

exact!

Now,

$$F(x, y) = \int \left(1 + \frac{y^2}{x^2} \right) dx$$

$$F(x, y) = x + y^2 \left(\frac{-1}{x} \right) + g(y)$$

$$F(x, y) = x - \frac{y^2}{x} + g(y) \quad \text{--- (1)}$$

Taking derivative wrt y on b/s:-

$$F_y = \frac{-2y}{x} + g'(y)$$

$$\frac{-2y}{x} = \frac{-2y}{x} + g'(y)$$

$$g'(y) = 0$$

$$g(y) = 0$$

eq (1) becomes:-

$$F(x, y) = x - \frac{y^2}{x}$$

q (1) answer:-

$$F(x, y) = x - \frac{y^2}{x}$$

$$C = x - \frac{y^2}{x}$$

8. $e^x(\cos y \, dx - \sin y \, dy) = 0$

$$M = e^x \cos y \quad N = -e^x \sin y$$

$$\frac{\partial M}{\partial y} = -e^x \sin y \quad \frac{\partial N}{\partial x} = -e^x \sin y$$

exact!

$$F(x, y) = \int e^x \cos y \cdot dx$$

$$F(x, y) = e^x \cdot \cos y + g(y) \quad \text{--- (i)}$$

$$F_y = e^x \cdot \sin y + g'(y)$$

$$-e^x \sin y = -e^x \sin y + g'(y)$$

$$g'(y) = 0$$

$$g(y) = 0$$

so eq (i)

$$F(x, y) = e^x \cos y$$

$$C = e^x \cos y$$

$$C = e^x \cos y$$

A. $e^{2x}(2 \cos y dx - \sin y dy) = 0, \quad y(0) = 0$

$$M = e^{2x} \cdot 2 \cos y \quad N = -e^{2x} \sin y$$

$$\frac{\partial M}{\partial y} = -2e^{2x} \sin y \quad \frac{\partial N}{\partial x} = -2e^{2x} \sin y$$

exact!

$$F(x, y) = \int e^{2x} \cdot 2 \cos y \cdot dx$$

$$F(x, y) = e^{2x} \cos y + g(y) \text{ --- (1)}$$

Taking derivative-

$$F_y = -e^{2x} \sin y + g'(y)$$

$$\cancel{-e^{2x} \sin y} = \cancel{-e^{2x} \sin y} + g'(y)$$

$$g'(y) = 0$$

$$g(y) = 0$$

- eq (1) becomes-

$$F(x, y) = e^{2x} \cos y$$

$$C = e^{2x} \cos y$$

giving condition: $y(0) = 0$

initial condition: $y(0) = 0$

$$C = e^{2u} \cos(0)$$

$$\boxed{C = 1}$$

So,

$$1 = e^{2u} \cos y$$

10. $y dx + [y + \tan(x+y)] dy = 0, \quad \cos(x+y)$

$$M = y$$

$$\frac{\partial M}{\partial y} = 1$$

$$N = y + \tan(xy)$$

$$\frac{\partial N}{\partial x} = \sec^2(xy)$$

Not - exact!

→ using provided integration factor.

$$M = \cos(xy) \cdot y$$

$$N = \cos(xy) \{ y + \tan(xy) \}$$

$$\frac{\partial M}{\partial y} = y \cdot -\sin(xy) + \cos(xy)$$

$$N = y \cos(xy) + \cos(xy) \cdot \tan(xy)$$

$$N = y \cos(xy) + \cancel{\cos(xy)} \cdot \frac{\sin(xy)}{\cancel{\cos(xy)}}$$

$$N = y \cos(xy) + \sin(xy)$$

$$\frac{\partial N}{\partial x} = \cos(xy)(1) + y \cdot -\sin(xy) + \cos(xy)$$

$$\frac{\partial N}{\partial x} = -y \sin(xy) + \cos(xy) = \frac{\partial M}{\partial y} \quad \text{exact!}$$

$$\rightarrow F(x, y) = \int y \cos(xy) \cdot dx$$

$$F(x, y) = y \cdot \sin(xy) + g(y) \quad \text{--- (1)}$$

- derivative wrt 'y' :-

$$F_y = \sin(xy) + y \cos(xy) + g'(y)$$

$$\cancel{y \cos(xy)} + \cancel{\sin(xy)} = \cancel{\sin(xy)} + y \cos(\cancel{xy}) + g'(y)$$

$$g'(y) = 0$$

$$g(y) = 0$$

eq (1) becomes:-

$$F(x, y) = y \sin(xy)$$

$$C = y \sin(xy)$$

16. TEAM PROJECT. Solution by Several Methods.

Show this as indicated. Compare the amount of work.

(a) $e^y(\sinh x dx + \cosh x dy) = 0$ as an exact ODE and by separation.

(b) $(1 + 2x) \cos y dx + dy/\cos y = 0$ by Theorem 2 and by separation.

(c) $(x^2 + y^2) dx - 2xy dy = 0$ by Theorem 1 or 2 and by separation with $v = y/x$.

(d) $3x^2 y dx + 4x^3 dy = 0$ by Theorems 1 and 2 and by separation.

(e) Search the text and the problems for further ODEs that can be solved by more than one of the methods discussed so far. Make a list of these ODEs. Find further cases of your own.

(a) exact ODE

$$e^y (\sinh x dx + \cosh x dy) = 0$$

$$M = e^y \sinh x$$

$$N = e^y \cosh x$$

$$\frac{\partial M}{\partial y} = e^y \sinh x$$

$$\frac{\partial N}{\partial x} = e^y \sinh x$$

$$F(x, y) = \int e^y \sinh x \cdot dx \quad \text{exact!}$$

$$F(x, y) = e^y \cdot \cosh x + g(y) \quad \text{--- (1)}$$

Taking derivative wrt y .

$$F_y = e^y \cdot \cosh x + g'(y)$$

$$0 = \dots \quad 0 = \dots \quad \text{--- (2)}$$

$$r_y = e^x \cdot \cosh x + g'(y)$$

$$e^x \cdot \cosh x = e^x \cosh x + g'(y)$$

$$g'(y) = 0$$

$$g(y) = 0$$

- eq ① becomes

$$F(x, y) = e^x \cosh x$$

$$C = e^x \cosh x$$

$$e^x (\sinh x \cdot dx + \cosh x \cdot dy) = 0$$

$$e^x \neq 0$$

$$\sinh x \cdot dx = -\cosh x \cdot dy$$

$$-\tanh x \cdot dx = dy$$

Taking $\int$ on b/s

$$-\ln |\cosh x| + C = y$$

$$C - \ln |\cosh x| = y$$

$$\ln |C| - \ln |\cosh x| = y$$

$$\ln \left| \frac{C}{\cosh x} \right| = y$$

$$\frac{C}{\cosh x} = e^y$$

$$C = e^y \cosh x$$

Both methods give same answer:

⑥ Thorm (2):-

$$(1+2x) \cos y \cdot dx + \frac{1}{\cos y} \cdot dy$$

$$(1+2x) \cos^2 y \cdot dx + 1 \cdot dy = 0 \quad \text{--- (a)}$$

will give non-exact :-

$$u(x) = e^{\int \frac{N_x - M_y}{M} \cdot dy}$$

∴ because the
variable y is
given / depends on

$$= e^{\int - \left\{ \frac{-2(1+2x) \sin y \cos y}{(1+2x) \cos^2 y} \right\}}$$

$$= e^{\int 2 \tan y \cdot dy}$$

$$= e^{-2 \ln |\cos y|}$$

$$= e^{(\cos y)^{-2}}$$

$$= \sec^2 y$$

- multiply with (a) :-

$$(1+2x) \cos^2 y \cdot \sec^2 y \cdot dy = 0$$

$$(1+2x) dx + \sec^2 y \cdot dy = 0$$

Now,

$$F(x, y) = \int 1+2x \cdot dx$$

$$F(x, y) = x + x^2 + g(y) \quad \text{--- (i)}$$

$$F(x, y) = x + x^2 + g(y) \quad \text{--- (1)}$$

$$F_y = g'(y)$$

$$\sec^2 y = g'(y)$$

$$g(y) = \tan(y)$$

$$F(x, y) = x + x^2 + \tan(y)$$

∴ implicit solution-

$$C = x + x^2 + \tan(y)$$

separable method-

using eq (1)

$$(1 + 2x) \cos^2 y \cdot dx + dy = 0$$

$$(1 + 2x) \cos^2 y \cdot dx = -dy$$

$$\int (1 + 2x) \cdot dx = \int -\sec^2 y \cdot dy$$

$$x + x^2 + C = -\tan(y)$$

$$C = -\tan y - x - x^2$$

$$\boxed{C = \tan y + x + x^2}$$

$$(c) \quad (x^2 + y^2) dx - 2xy \cdot dy = 0$$

$$M = x^2 + y^2$$

$$N = -2xy$$

$$\frac{\partial M}{\partial y} = 2y$$

$$\frac{\partial N}{\partial x} = -2y$$

Not - exact.

$$u(x) = e \int \frac{2xy}{-2xy} \cdot dx$$

$$= e \int -\frac{2}{x} \cdot dx$$

$$\boxed{= 1/x^2}$$

$$\frac{x^2 + y^2}{x^2} \cdot dx - \frac{2xy}{x^2} \cdot dy = 0$$

$$\left(1 + \frac{y^2}{x^2}\right) \cdot dx - \frac{2y}{x} \cdot dy = 0$$

$$F(x, y) = \int \left(1 + \frac{y^2}{x^2}\right) \cdot dx$$

$$= x - \frac{y^2}{x} + g(y)$$

$$F_y = -\frac{2y}{x} + g'(y)$$

$$\cancel{\frac{-2y}{x}} = \cancel{\frac{-2y}{x}} + g'(y)$$

$$g'(y) = 0$$

$$g(y) = 0$$

So,

$$F(x, y) = x - \frac{y^2}{x}$$

$$C = x - \frac{y^2}{x}$$

separation by $y = vx$

$$y = vx$$

$$\frac{dy}{dx} = x \cdot \frac{dv}{dx} + v$$

from the given ODE:-

$$\frac{dy}{dx} = \frac{x^2 + y^2}{2xy}$$

$$x \cdot \frac{dv}{dx} + v = \frac{x^2 + x^2 v^2}{2xvx}$$

$$x \frac{dv}{dx} + v = \frac{1 + v^2}{2v}$$

$$\int \frac{1}{x} \cdot dx = \int \frac{2v}{1 - v^2} \cdot dv$$

$$\ln|x| + C = -\ln|1 - v^2|$$

$$\ln|1 - v^2| = -\ln|x| + C$$

$$1 - v^2 = \frac{C}{x}$$

$$1 - \frac{y^2}{x^2} = \frac{C}{x}$$

$$C = x - \frac{y^2}{x}$$

$$d \quad 3x^2y \cdot dx + 4x^3 \cdot dy = 0$$

$$M = 3x^2y \quad N = 4x^3$$

$$\frac{\partial M}{\partial y} = 3x^2 \quad \frac{\partial N}{\partial x} = 12x^2$$

Non-Exact!

$$u(x) = e^{\int \frac{3x^2 - 12x^2}{4x^3} \cdot dx}$$

$$= e^{\int \frac{(-9)}{4x} \cdot dx}$$

$$= e \cdot \frac{-9}{4} \cdot \ln(x)$$

$$= e^{\ln|x| - 9/4}$$

$$= x^{-9/4}$$

$$u(y) = e^{\int \frac{12x^2 - 3x^2}{3x^2y} \cdot dy}$$

$$u(y) = e^{\int \frac{9x^2}{3x^2y} \cdot dy}$$

$$= e^{3\ln(y)}$$

$$= y^3$$

1. (2, 1)

2. (1, 1)

1

1. (2, 1)

2. (2, 1)

$$F(x, y) = \int 3x^{-1/4} y \cdot dx$$

$$= 4y x^{3/4} \cdot g(y)$$

$$F_y = 4x^{3/4} + g'(y)$$

$$4x^{3/4} = 4x^{3/4} + g'(y)$$

$$g'(y) = 0$$

$$g(y) = 0$$

$$F(x, y) = \int 3x^2 y^4 \cdot dy$$

$$= y^5 x^3 + g(y)$$

$$F_y = 4x^3 y^3 + g'(y)$$

$$4x^3 y^3 = 4x^3 y^3 + g'(y)$$

,

$$g'(y) = 0$$

$$g(y) = 0$$

So,

$$F(x, y) = 4y x^{3/4}$$

$$C = 4y x^{3/4}$$

$$\boxed{C = y x^{3/4}}$$

So,

$$F(x, y) = y^5 x^3$$

$$\frac{C}{x^3} = y^5$$

$$y = \frac{C^{1/5}}{x^{3/5}}$$

same

$$\boxed{y x^{3/5} = C}$$

Separable method-

$$4x^3 dy = -3x^2 y \cdot dx$$

$$\frac{dy}{dx} = \frac{-3y}{4x}$$

$$\int \frac{1}{y} \cdot dy = \int \frac{-3}{4x} \cdot dx$$

$$\int \frac{1}{y} \cdot dy = \int \frac{-3}{4x} \cdot dx$$

$$\ln|y| = \frac{-3}{4} \ln|x| + C$$

$$\ln|y| = \ln|Cx^{-3/4}|$$

$$\boxed{C = yx^{3/4}}$$

SECTION 1.5:

$$5. \quad y' + ky = e^{-kx}$$

$$\begin{aligned} \mu(x) &= e^{\int k \cdot dx} \\ &= e^{kx} \end{aligned}$$

So,

$$e^{kx} y' + ke^{kx} y \cdot e^{-kx} \cdot e^{-kx}$$

$$\frac{d}{dx} (e^{kx} y) = x + C$$

$$y = \frac{x}{e^{kx}} + C$$

$$6. \quad y' + 2y = 4 \cos 2x, \quad y(\frac{1}{4}\pi) = 3$$

$$\mu(x) = e^{\int 2 \cdot dx}$$

$$y(x) = e^{\int 2 \cdot dx}$$

$$= e^{2x}$$

$$e^{2x} y' + 2e^{2x} y = e^{2x} \cdot 4 \cos 2x$$

$$\int \frac{d}{dx} (e^{2x} y) \cdot dx = \int e^{2x} \cdot 4 \cos 2x \, dx$$

$$e^{2x} y = e^{2x} (\cos 2x + \sin 2x) + C$$

$$y = \cos 2x + \sin 2x + C$$

$$1. \quad xy' = 2y + x^3 e^x$$

$$xy' - 2y = x^3 e^x$$

$$y' - \frac{2}{x} y = x^2 e^x$$

$$y(x) = e^{\int \frac{-2}{x}}$$

$$= e^{-2 \ln(x)}$$

$$= \frac{1}{x^2}$$

$$\frac{y'}{x^2} - \frac{2}{x^2} y = e^x$$

$$\int \frac{d}{dx} (x^{-2} y) = \int e^x \, dx$$

$$x^{-2} y = e^x + C$$

$$x^{-2}y = e^{-x} + C$$

$$y = x^2 e^x + C$$

11. $y' = (y - 2) \cot x$

$$\frac{dy}{dx} = (y - 2) \cot x$$

$$\int \frac{1}{y-2} \cdot dy = \int \cot x \cdot dx$$

$$\ln|y-2| = \ln|\sin x| + C$$

$$y = 2 + C \sin x$$

10. TEAM PROJECT. Riccati Equation. Clairaut Equation. Singular Solution.

A **Riccati equation** is of the form

$$(14) \quad y' + p(x)y = g(x)y^2 + h(x).$$

A **Clairaut equation** is of the form

$$(15) \quad y = xy' + g(y').$$

(a) Apply the transformation $y = Y + 1/u$ to the Riccati equation (14), where Y is a solution of (14), and obtain for u the linear ODE $u' + (2Yg - p)u = -g$. Explain the effect of the transformation by writing it as $y = Y + v$, $v = 1/u$.

(b) Show that $y = Y = x$ is a solution of the ODE $y' - (2x^3 + 1)y = -x^2y^2 - x^4 - x + 1$ and solve this Riccati equation, showing the details.

(c) Solve the Clairaut equation $y'^2 - xy' + y = 0$ as follows. Differentiate it with respect to x , obtaining

(b) Show that $y = Y = x$ is a solution of the ODE $y' - (2x^3 + 1)y = -x^2y^2 - x^4 - x + 1$ and solve this Riccati equation, showing the details.

(c) Solve the Clairaut equation $y'^2 - xy' + y = 0$ as follows. Differentiate it with respect to x , obtaining $y''(2y' - x) = 0$. Then solve (A) $y'' = 0$ and (B) $2y' - x = 0$ separately and substitute the two solutions (a) and (b) of (A) and (B) into the given ODE. Thus obtain (a) a general solution (straight lines) and (b) a parabola for which those lines (a) are tangents (Fig. 6 in Prob. Set 1.1); so (b) is the envelope of (a). Such a solution (b) that cannot be obtained from a general solution is called a **singular solution**.

(d) Show that the Clairaut equation (15) has as solutions a family of straight lines $y = cx + g(c)$ and a singular solution determined by $g'(s) = -x$, where $s = y'$, that forms the envelope of that family.

$$\textcircled{a} \quad y' + p(x)y = g(x)y^2 + h(x)$$

let Y be the particular solution then,

$$y = Y + \frac{1}{u}$$

∴

$$y' = Y' - \frac{u'}{u^2}$$

equation becomes

$$Y' - \frac{u'}{u^2} + p\left(Y + \frac{1}{u}\right) = g\left(Y + \frac{1}{u}\right) + h$$

Since Y is a solution $Y' + pY = gY^2 + h$

$$\cancel{y'} + \cancel{p}y - \frac{u'}{u^2} + \frac{p}{u} = \cancel{g}y^2 + \frac{2gy}{u} + \frac{g}{u^2} + \cancel{h}$$

$$\frac{-u'}{u^2} + \frac{p}{u} = \frac{2gy}{u} + \frac{g}{u^2}$$

$$-u' + pu = 2gyu + g$$

$$-g = u' + (2gy - p)u$$

b) $y' - (2x^3 + 1)y = -x^2y^2 - x^4 - x + 1$

$y = x$ is a solution

$$y' = 1$$

$$1 - (2x^3 + 1)x = -x^2x^2 - x^4 - x + 1$$

$$1 - 2x^4 - x = 1 - 2x^4 - x \text{ is a solution.}$$

So,

$$p(x) = -2(x^3 + 1), \quad g(x) = -x^2, \quad h(x) = -x^4 - x + 1$$

we put (a) :-

$$y = \frac{1}{u}$$

$$= x + \frac{1}{u}$$

- for verification

$$2gy - p = 2(-x^2)(x) - \{- (2x^3 + 1)\}$$

$$2gy - p = 2(-x^2)(x) - \{-(2x^2+1)\}$$

$$= 1$$

$$-q = x^2$$

So,

$$u' + u = x^2$$

$$u(x) = \exp. \int 1. du$$

$$= e^x$$

$$e^x u' + e^x u = e^x x^2$$

$$\int e^x u = \int e^x x^2$$

$$e^x u = e^x (x^2 - 2x + 2) + C$$

$$u = x^2 - 2x + 2 + Ce^{-x}$$

hence,

$$y(x) = x + \frac{1}{x^2 - 2x + 2 + Ce^{-x}}$$

(C) $(y')^2 - xy' + y = 0$ — (A)

Taking derivative

$$2y' y'' - y' - xy'' + y' = 0$$

$$2y' y'' - xy'' = 0$$

$$y''(2y' - x) = 0$$

$$y'' = 0 \quad \text{or} \quad 2xy' - x = 0$$

if $y=0$:- $y'=c$ $y=cx+d$

$$c - xc + cx + d = 0$$

$$c^2 + d = 0$$

$$d = -c^2$$

$$\boxed{y = cx - c^2}$$

• If $2y - x = 0$, then $y = \frac{x}{2}$ $\Rightarrow y = \frac{x^2}{4} + k$

Prngy (A)

$$\left(\frac{x}{2}\right)^2 - x\left(\frac{x}{2}\right) + \left(\frac{x^2}{4} + k\right) = k$$

$$\frac{x^2}{4} - \frac{x^2}{2} + \frac{x^2}{4} + k \Rightarrow k$$

Now for (A) to hold

by $y = \frac{x^2}{4}$ $\therefore k=0$

(d)

$$y = xy' + g(y')$$

let $v = y'$

$$y = xv + g(v)$$

Taking derivative wrt x on b/s:-

$$v' = v + x v' + g'(v) v'$$

taking derivative wrt x on b/s:-

$$y' = v + xv' + g'(v)v'$$

$$0 = \int (x + g'(v)) v'$$

$$v' = 0, \quad x = g'(v) = 0$$

- If $v' = 0$ then $v = c$
 $y = xc + g(c)$

- If $x + g'(v) = 0$ then $g'(v) = -x$
 $y = xv + g(v)$

General solution $y = cx + g(c)$.

singular solution $g'(s) = -x$

31. Newton's law of cooling. If the temperature of a cake is 300°F when it leaves the oven and is 200°F ten minutes later, when will it be practically equal to the room temperature of 60°F , say, when will it be 61°F ?

$$\begin{aligned} T' &= k(T - T_r) \\ &= k(T - 60) \end{aligned}$$

where,

$$T(t) - 60 = Ce^{kt}$$

at $t = 0$; $T(0) = 300$

$$300 - 60 = Ce^{k(0)}$$

$$300 - 60 = C e^{k(0)}$$

$$C = 240$$

So,

$$T(t) - 60 = 240 e^{kt}$$

At $t = 10$ -

$$T = 200$$

$$200 - 60 = 240 e^{kt}$$

$$140 = 240 e^{10k}$$

$$k = \frac{1}{10} \ln\left(\frac{7}{12}\right)$$

We need $T(t) = 61$ at specific t

$$61 - 60 = 240 e^{(1/10 \ln(7/12))t}$$

$$\frac{1}{240} = e^{(1/10 \ln(7/12))t}$$

$$t = 10 \frac{\ln(1/240)}{\ln(7/12)}$$

$$\boxed{t \approx 102 \text{ min}}$$

33. Drug injection. Find and solve the model for drug injection into the bloodstream if, beginning at $t = 0$, a constant amount A g/min is injected and the drug is simultaneously removed at a rate proportional to the amount of the drug present at time t .

$$Q' = A - kQ$$

$$Q' + kQ = A$$

$$p(x) = e^{\int k \cdot dx} = e^{kx}$$

$$e^{kx} Q' + k e^{kx} Q = e^{kx} A$$

$$\int \frac{d}{dx} (e^{kx})(Q) = \int e^{kx} A \cdot dx$$

$$Q e^{kx} = \frac{A}{k} e^{kx} + C$$

$$Q = \frac{A}{k} + C e^{-kx}$$

initial at $x=0$, $Q=0$

$$Q(0) = \frac{A}{k} + C$$

$$C = -\frac{A}{k}$$

So,

$$Q(x) = A \left(1 - e^{-kx} \right)$$

So,

$$Q(x) = \frac{A}{K} (1 - e^{-Kx})$$

25. Lake Erie. Lake Erie has a water volume of about 450 km^3 and a flow rate (in and out) of about 175 km^2

per year. If at some instant the lake has pollution concentration $p = 0.04\%$, how long, approximately, will it take to decrease it to $p/2$, assuming that the inflow is much cleaner, say, it has pollution concentration $p/4$, and the mixture is uniform (an assumption that is only imperfectly true)? First guess.

$$C(t) = C_{in} + (C_0 - C_{in}) e^{-Kt}$$

C_0 = initial conc.

C_{in} = inwards conc.

$$C(t) = \frac{p}{4} + \frac{3p}{4} e^{-Kt}$$

requirement is $C(t)$ of $p/2$

$$\frac{p}{2} = \frac{p}{4} + \frac{3p}{4} e^{-Kt}$$

$$e^{-Kt} = \frac{1}{3}$$

So,

$$t = \frac{1}{K} \ln(3)$$

Now, water volume is 450 km^3 flow rate of 175 km^2

Now, water volume is 450 km^3 flow rate of 175 km^3 both are constant.

$$k = \frac{r}{v} = \frac{175}{450}$$

$$t = \frac{450 \ln(3)}{175}$$

$$t \approx 2.8 \text{ years}$$

SECTION 2.2:

$$2. y'' + 36y = 0$$

$$\text{let } y = e^{mx}$$

$$m^2 + 36 = 0$$

$$m^2 = -36$$

$$m = \pm 6i$$

general solution:-

$$m = \alpha \pm \beta i$$

$$y(x) = e^{\alpha x} (A \cos(\beta x) + B \sin(\beta x))$$

$$\text{Here, } \alpha = 0, \beta = 6$$

$$y(x) = A \cos(6x) + B \sin(6x)$$

$$y'(x) = -6A \sin 6x + 6B \cos 6x$$

$$\begin{aligned} y''(x) &= -36A \cos 6x - 36B \sin 6x \\ &= -36y \end{aligned}$$

$$y''(x) = -36 \cos 6x - 36 \sin 6x$$

$$= -36y$$

$$y'' + 36y = -36y + 36y$$

$$= 0$$

so,

$$y = A \cos 6x + B \sin 6x \text{ is the solution}$$

$$4. y'' + 4y' + (\pi^2 + 4)y = 0$$

Characteristic equation-

$$m^2 + 4m + (\pi^2 + 4) = 0$$

$$m = \frac{-4 \pm \sqrt{16 - 4(\pi^2 + 4)}}{2}$$

$$m = -2 \pm i\pi$$

for

$$m = \alpha + i\beta \text{ with } \alpha = -2 \text{ and } \beta = \pi$$

$$y(x) = e^{-2x} (A \cos(\pi x) + B \sin(\pi x)) \text{ Ans}$$

$$12. y'' + 9y' + 20y = 0$$

Characteristic equation-

$$m^2 + 9m + 20 = 0$$

$$m = \frac{-9 \pm \sqrt{81 - 80}}{2}$$

$$m^2 + 9m + 20 = 0$$

$$m = \frac{-9 \pm \sqrt{81 - 4(20)}}{2}$$

$$m = \frac{-9 \pm 1}{2}$$

$$m = -4, -5$$

$$y(x) = C_1 e^{-4x} + C_2 e^{-5x}$$

by substitution-

$$\text{let } y = e^{-4x}$$

$$y' = -4e^{-4x}$$

$$y'' = 16e^{-4x}$$

$$y'' + 9y' + 20y = 16e^{-4x} + 9(-4e^{-4x}) + 20e^{-4x}$$

$$= 16 - 36 + 20 = 0$$

Now, e^{-5x}

$$y' = -5e^{-5x}$$

$$y'' = 25e^{-5x} = 25 - 45 + 20 = 0$$

$$14. y'' + 2k^2 y' + k^4 y = 0$$

Equation-

$$m^2 + 2k^2 m + k^4 = 0$$

$$m = \frac{-2k^2 \pm \sqrt{4k^4 - 4k^4}}{2}$$

$$m = -k^2 \quad (\text{double root})$$

$$\boxed{y(x) = (c_1 + c_2 x) e^{-k^2 x}} \quad \text{Ans}$$

18. $\cos 2\pi x, \sin 2\pi x$

given basis

$$\cos 2\pi x, \sin 2\pi x$$

general solution

$$y(x) = A \cos(2\pi x) + B \sin(2\pi x)$$

characteristic equation from roots

$$(m - 2\pi i)(m + 2\pi i) = 0$$

$$m^2 + (2\pi)^2 = 0$$

$$m^2 + 4\pi^2 = 0$$

from $m^2 + 4\pi^2 y = 0$ ODE is $y'' + 4\pi^2 y = 0$

$$a = 0, \quad b = 4\pi^2$$

$$y = \cos(2\pi x)$$

$$y' = -2\pi \sin(2\pi x)$$

$$y'' = -4\pi^2 \cos(2\pi x)$$

$$y'' + 4\pi^2 y = -4\pi^2 \cos(2\pi x) + 4\pi^2 \cos(2\pi x) \\ = 0$$

So, for $\sin x$:-

$$y'' + 4\pi^2 y = 0$$

$$19. e^{(-2+i)x}, e^{(-2-i)x}$$

$$m_1 = -2+i, m_2 = -2-i$$

Characteristic equation:-

$$m_1 + m_2 = -2+i - 2-i \\ = -4$$

$$m_1 m_2 = (-2+i)(-2-i) \\ = 4 + 2i - 2i - i^2 \\ = 5$$

So,

$$m^2 - (m_1 + m_2)m + m_1 m_2 = 0$$

$$m^2 - (-4)m + 5 = 0$$

$$m^2 + 4m + 5 = 0$$

$$a=4, b=5$$

Now check:-

$$\text{let } y = e^{(-2+i)x}$$

$$\text{let } y = e^{(-2+i)x}$$

$$y' = (-2+i)y, \quad y'' = (-2+i)^2 y$$

$$\begin{aligned} (-2+i)^2 &= 4 - 4i + i^2 = 4 - 4i - 1 \\ &= 3 - 4i \end{aligned}$$

Now,

$$\begin{aligned} y'' + 5y' + 4y &= [(3-4i) + 4(-2+i) + 5]y \\ &= 3 - 4i - 8 + 4i + 5 \\ &= 0 \end{aligned}$$

similarly for m_2

$$[y'' + 4y' + 5y = 0] \text{ then}$$

SECTION 2.7

Find a (real) general solution. State which rule you are using. Show each step of your work.

$$1. \quad y'' + 5y' + 4y = 10e^{-3x}$$

• y_h

$$y_h'' + 5y_h' + 4y_h = 0$$

$$m^2 + 5m + 4 = 0$$

$$m = -4, -1$$

$$y_h = C_1 e^{-4x} + C_2 e^{-x}$$

Using methods of undetermined coefficients,

$$y_h = Ae^{-3x}$$

Then

$$y_p' = -3Ae^{-3x}, \quad y_p'' = 9Ae^{-3x}$$

$$y'' + 5y' + 4y = 9Ae^{-3x} + 5(-3Ae^{-3x}) + 4Ae^{-3x} \\ = -2Ae^{-3x}$$

Now set equal

$$\cancel{10e^{-3x}} = \cancel{-2Ae^{-3x}} \\ \boxed{A = -5}$$

So,

$$y_p = -5e^{-3x}$$

general solution.

$$y = y_h + y_p \\ \boxed{y = C_1 e^{-x} + C_2 e^{-2x} - 5e^{-3x}}$$

2. $10y'' + 50y' + 57.6y = \cos x$

divide by 10:-

$$y'' + 5y' + 5.76y = \frac{\cos x}{10}$$

$$y_h'' + 5y_h' + 5.76y_h = 0$$

$$m^2 + 5m + 5.76 = 0$$

$$m = -1.2 \quad m = -3.2$$

$$m^2 + 5m + 5.76 = 0$$

$$m_1 = 1.8, \quad m_2 = -3.2$$

$$y_h = C_1 e^{-1.8x} + C_2 e^{-3.2x}$$

$$y_p = k \cos x + M \sin x$$

$$y'_p = -k \sin x + M \cos x$$

$$y''_p = -k \cos x - M \sin x$$

• After substitution:-

$$(4.76k + 5M) \cos x + (-5k + 4.76M) \sin x = 0.1 \cos x + 0$$

solve for k and M

$$-5k + 4.76M = 0$$

$$M = \frac{5k}{4.76}$$

$$4.76^2 k + 25k = 4.76 \times 0.1$$

$$k = 0.01$$

$$M = 0.01$$

general solution:-

$$y = C_1 e^{-1.8x} + C_2 e^{-3.2x} + 0.01 \cos x + 0.01 \sin x$$

$$5. \quad y'' + 4y' + 4y = e^{-x} \cos x$$

$$y'' + 4y' + 4y = 0$$

characteristic eqn:-

$$m^2 + 4m + 4 = 0$$

$$m = -2$$

$$y_h = C_1 e^{-2x} + C_2 x e^{-2x}$$

for particular solution-

$$y_p = e^x (A \cos x + B \sin x)$$

$$y_p' = e^{-x} [(-A+B) \cos x + (-B-A) \sin x]$$

$$y_p'' = e^{-x} [(-A+B) \cos x + (-B-A) \sin x] + e^{-x} [(-A+B) \sin x + (-B-A) \cos x]$$

$$\text{ODE} = y_p'' + 4y_p' + 4y_p = e^{-x} \cos x$$

$$-2B + (-4A + 4B) + 4A = 0 + 2B$$

$$y_p'' + 4y_p' + 4y_p = e^{-x} [2B \cos x - 2A \sin x]$$

$$2B = 1 \quad \boxed{B = 1/2} \quad \boxed{A = 0}$$

general solution-

$$y(x) = C_1 e^{-2x} + C_2 x e^{-2x} + 0.5 e^{-x} \sin x$$

$$6. y'' + y' + (\pi^2 + \frac{1}{4})y = e^{-x/2} \sin \pi x$$

solving for homogeneous

$$y'' + y' + (\pi^2 + \frac{1}{4})y = 0$$

characteristic

$$m^2 + m + (\pi^2 + \frac{1}{4}) = 0$$

$$m = -\frac{1}{2} \pm \pi i$$

$$y_h = e^{-x/2} (C_1 \cos(\pi x) + C_2 \sin(\pi x))$$

Particular solution-

$$y_p = x e^{-x/2} (A \cos(\pi x) + B \sin(\pi x))$$

after solving-

$$y'_p = e^{-x/2} [A \cos(\pi x) + B \sin(\pi x)] + x e^{-x/2} [P \cos \pi x + Q \sin(\pi x)]$$

$$P = -\frac{1}{2} A + \pi B \quad Q = -\frac{1}{2} B - \pi A$$

general solution-

$$y = e^{-x/2} \left[C_1 \cos(\pi x) + C_2 \sin(\pi x) - \frac{x}{2\pi} \cos(\pi x) \right] \text{ h.c.}$$

14. $y'' + 4y' + 4y = e^{-2x} \sin 2x, \quad y(0) = 1,$
 $y'(0) = -1.5$

Solving homo eqn

$$y'' + 4y' + 4y = 0$$

$$m^2 + 4m + 4 = 0$$

$$m = -2$$

$$y_h = C_1 e^{-2x} + C_2 x e^{-2x}$$

Particular solution-

$$y_p = A x^2 e^{-2x}$$

$$y'_p = A e^{-2x} (2x - 2x^2)$$

$$y''_p = A e^{-2x} [2 - 8x + 4x^2]$$

Put into ODE:-

$$y''_p + 4y'_p + 4y_p = A e^{-2x} (2)$$

set equal:-

$$\frac{2A e^{-2x}}{A = 0.5}$$

$$y_p = \frac{1}{2} x^2 e^{-2x}$$

$$y = C_1 e^{-2x} + C_2 x e^{-2x} + \frac{1}{2} x^2 e^{-2x}$$

$$y = C_1 e^{-2x} + C_2 x e^{-2x} + \frac{1}{2} x^2 e^{-2x}$$

$$y(0) = 1$$

$$1 = C_1 + 0 + 0$$

$$C_1 = 1$$

Now

$$y' = e^{-2x} [-2C_1 + C_2 - 2C_2 x + x - x^2]$$

$$y'(0) = 1.5$$

$$-\frac{3}{2} = -2(1) + C_2$$

$$C_2 = 0.5$$

Particular solution

$$[y = e^{-2x} (1 + 0.5x + 0.5x^2)] \text{ is -}$$

$$15. (x^2 D^2 - 3xD + 3I)y = 3 \ln x - 4, \\ y(1) = 0, \quad y'(1) = 1; \quad y_p = \ln x$$

$$D = \frac{d}{dx}, \quad I \text{ is identity operator}$$

$$x^2 y'' - 3xy' + 3y = 3 \ln(x) - 4$$

Homog. eq.:-

$$m(m-1) - 3m + 3 = 0$$

$$m^2 - 4m + 3 = 0$$

$$m = 1, 3$$

$$m^2 - 4m + 3 = 0$$

$$(m-1)(m-3) = 0$$

$$m_1 = 1, m_2 = 3$$

$$y_h = C_1 x + C_2 x^3$$

Now

$$y_p = \ln x$$

$$y_p' = \frac{1}{x} \quad y_p'' = -\frac{1}{x^2}$$

$$x^2 \left(-\frac{1}{x^2} \right) - 3x^2 \left(\frac{1}{x} \right) + 3 \ln x$$

$$= -4 + 3 \ln x$$

General

$$y(x) = C_1 x + C_2 x^3 + \ln(x)$$

$$y(1) = 0$$

$$C_1 + C_2 = 0$$

or

$$y'(x) = C_1 + 3C_2 x^2 + \frac{1}{x}$$

$$y'(1) = C_1 + 3C_2 + 1$$

$$0 = C_1 + 3C_2$$

so,

$$C_2 = 0, C_1 = 0$$

$$C_2 = 0, C_1 = 0$$

So,

$$[y(u) = \ln(u)] \text{ Ans}$$

SECTION 2.9:

- 1. RC-Circuit.** Model the RC-circuit in Fig. 64. Find the current due to a constant E .

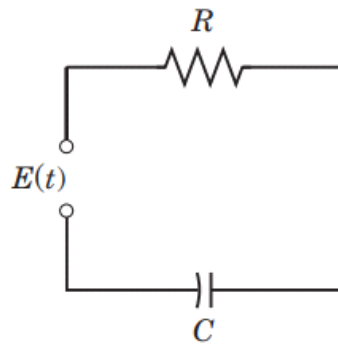


Fig. 64. RC-circuit

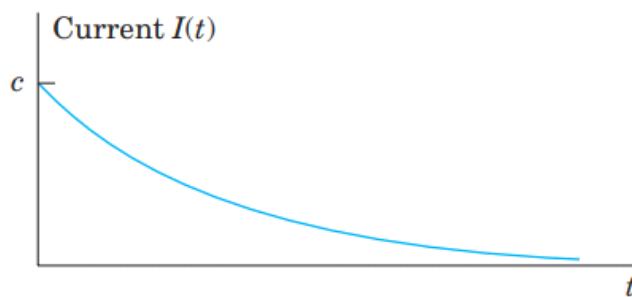


Fig. 65. Current I in Problem 1

Very nice

$$E = V_R + V_C$$

$$V_R = RI(t)$$

$$\dots$$

$$V_R = RI(t)$$

$$V_C = \frac{1}{C} \int I(t) \cdot dt \Rightarrow I = C \frac{dV_C}{dt}$$

$$E = R \frac{dQ}{dt} + \frac{Q}{C}$$

$$R \frac{dQ}{dt} + \frac{Q}{C} = E$$

$$u(t) = e^{\int 1/RC dt}$$

$$= e^{t/RC}$$

multiply by $e^{t/RC}$

$$e^{t/RC} \frac{dQ}{dt} + \frac{1}{RC} e^{t/RC} Q = \frac{E}{R} e^{t/RC}$$

$$e^{t/RC} \cdot Q = EC e^{t/RC} + k$$

$$Q(t) = EC + k e^{-t/RC}$$

$$Q(0) = 0$$

$$0 = EC + k$$

$$k = -EC$$

so,

$$Q(t) = EC(1 - e^{-t/RC})$$

hence $I(t)$

$$I(t) = \frac{E}{R} e^{-t/RC} \quad \text{Ans}$$

$$\left[I(t) = \frac{E}{R} e^{-t/RC} \right] \text{ Ans}$$

3. **RL-Circuit.** Model the RL -circuit in Fig. 66. Find a general solution when R, L, E are any constants. Graph or sketch solutions when $L = 0.25$ H, $R = 10 \Omega$, and $E = 48$ V.

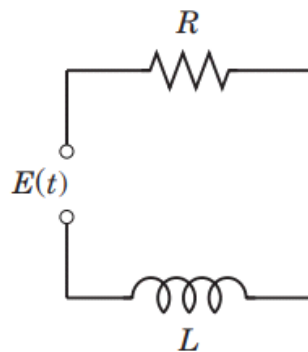


Fig. 66. RL -circuit

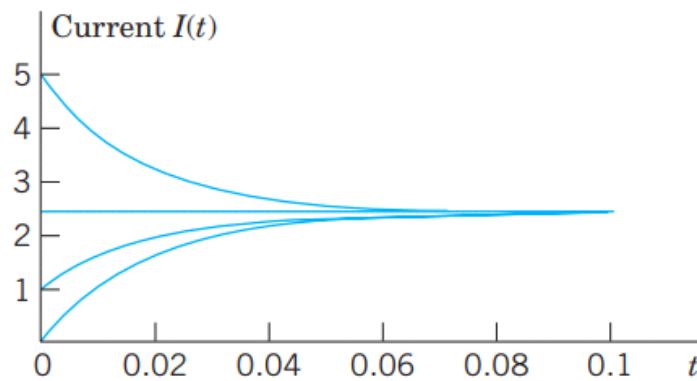


Fig. 67. Currents in Problem 3

using KVL

$$E = V_R + V_L$$

$$V_R = IR$$

$$V_L = L \frac{dI}{dt}$$

$$L \frac{dI}{dt} + IR = E$$

$$\frac{E}{L} = \frac{dI}{dt} + I \frac{R}{L}$$

$$y(t) = e^{\int R/L dt}$$

$$= e^{\frac{Rt}{L}}$$

Now

$$e^{\frac{Rt}{L}} \frac{dI}{dt} + \frac{R}{L} e^{\frac{Rt}{L}} I = \frac{E}{L} e^{\frac{Rt}{L}}$$

$$\frac{d}{dt} (e^{\frac{Rt}{L}} \cdot I) = \frac{E}{L} e^{\frac{Rt}{L}}$$

$$I(t) = ER + ke^{\frac{Rt}{L}}$$

given cases

$$L = 0.25H, R = 10\Omega, E = 48V$$

$$\frac{E}{R} = 4.8A, \quad \frac{R}{L} = \frac{10}{0.25} = 40s^{-1}$$

$$I(t) = 4.8 + ke^{-40t}$$

$$\text{At } I(0) = 0$$

$$0 = 4.8 + k$$

$$I(t) = 4.8(1 - e^{-40t})$$

$$v = 4.8 + k$$

$$I(t) = 4.8(1 - e^{-4t})$$

4. **RL-Circuit.** Solve Prob. 3 when $E = E_0 \sin \omega t$ and R , L , E_0 , and ω are arbitrary. Sketch a typical solution.

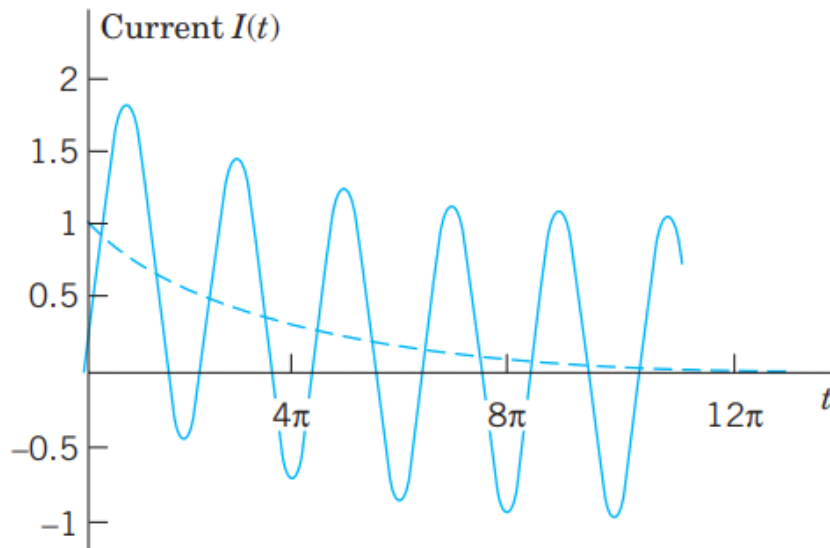


Fig. 68. Typical current $I = e^{-0.1t} + \sin(t - \frac{1}{4}\pi)$ in Problem 4

Very well.

$$L \frac{di}{dt} + Ri = E_0 \sin(\omega t)$$

$$\frac{dI}{dt} + \frac{R}{L} I = \frac{E_0}{L} \sin(\omega t)$$

let

$$a = \frac{R}{L}, \quad k = \frac{E_0}{L}$$

$$I_h(t) = Ce^{-at}$$

$$z_h(t) = C e^{-at}$$

Particular solution-

$$z_p = M \cos(\omega t) + N \sin(\omega t)$$

$$z_p' = a z_p = k \sin(\omega t)$$

$$z_p' = -\omega M \sin(\omega t) + \omega N \cos(\omega t)$$

$$[-\omega M \sin(\omega t) + \omega N \cos(\omega t)] + a [M \cos(\omega t) + N \sin(\omega t)] = k \sin \omega t$$

$$\omega N + a M = 0$$

$$-\omega M + a N = k$$

$$M = \frac{-\omega N}{a}$$

$$N = \frac{ak}{\omega^2 + a^2} \quad , \quad M = \frac{-\omega k}{\omega^2 + a^2}$$

$$\omega^2 + a^2 = \omega^2 + \frac{R^2}{L^2}$$

$$N = \frac{R E_0}{\omega^2 L^2 + R^2}$$

$$M = \frac{-\omega E_0 L}{\omega^2 L^2 + R^2}$$

So,

$$I_p(t) = \frac{E_0}{\omega^2 L^2 + R^2} [-\omega (\cos \omega t + R \sin \omega t)]$$

general solution-

$$I(t) = (e^{-\lambda t} + \frac{E_0}{z} \sin(\omega t - \phi)),$$

$$z = \sqrt{R^2 + (\omega L)^2}, \quad \phi = \tan^{-1}\left(\frac{\omega L}{R}\right)$$

8-14

Find the **steady-state current** in the *RLC*-circuit in Fig. 61 for the given data. Show the details of your work.

8. $R = 4 \, \Omega$, $L = 0.5 \, \text{H}$, $C = 0.1 \, \text{F}$, $E = 500 \sin 2t \, \text{V}$

9. $R = 4 \, \Omega$, $L = 0.1 \, \text{H}$, $C = 0.05 \, \text{F}$, $E = 110 \, \text{V}$

10. $R = 2 \, \Omega$, $L = 1 \, \text{H}$, $C = \frac{1}{20} \, \text{F}$, $E = 157 \sin 3t \, \text{V}$

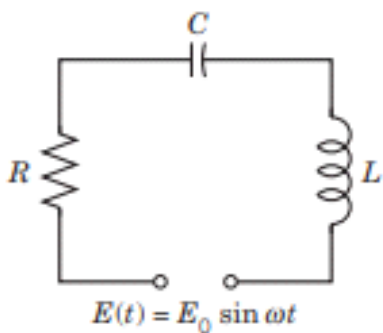


Fig. 61. *RLC*-circuit

general ODE we use KVL is

$$L I'' + R I' + \frac{1}{C} I = E'(t)$$

for sinusoidal:- $E(t) = E_0 \sin(\omega t)$

for sinusoidal:- $E(t) = E_0 \sin(\omega t)$

$$I_p(t) = \frac{E_0}{Z} \sin(\omega t - \phi)$$

here,

$$Z = R + j \left(\omega L - \frac{1}{\omega C} \right), \quad |Z| = \sqrt{R^2 + \left(\omega L - \frac{1}{\omega C} \right)^2}$$

$$\phi = \tan^{-1} \left(\frac{\omega L - 1/\omega C}{R} \right)$$

8) $R = 4 \Omega$ $L = 0.5 \text{ H}$ $C = 0.1 \text{ F}$ $E = 500 \sin 2t \text{ V}$
 $\omega = 2$

$$\omega L = 1$$

$$\frac{1}{\omega C} = 5$$

$$X_L - X_C = 1 - 5 = -4$$

$$|Z| = \sqrt{4^2 + 4^2} = 4\sqrt{2} \Omega$$

$$\phi = \tan^{-1} \left(\frac{-4}{4} \right) = -\frac{\pi}{4}$$

$$I_0 = \frac{500}{4\sqrt{2}} = 86.39 \text{ A}$$

$$I(t) = \frac{125}{\sqrt{2}} \sin \left(2t + \frac{\pi}{4} \right)$$

$$i(t) = \frac{125}{\sqrt{2}} \sin\left(2t + \frac{\pi}{4}\right)$$

9) $R=4 \quad L=0.1H \quad C=0.05F \quad E=110V$

Capacitor acts as open circuit $I_p=0$

10) $R=2\Omega, L=1H, C=\frac{1}{20}F$

$E=157 \sin 3tV \Rightarrow \omega=3$

$\omega L=3$

$$\frac{1}{\omega C} = \frac{1}{3\left(\frac{1}{20}\right)} = \frac{20}{3} = 6.667$$

$$X_L - X_C = 3 - \frac{20}{3} = -\frac{11}{3}$$

$$|Z| = \sqrt{2^2 + \left(-\frac{11}{3}\right)^2} = \sqrt{\frac{157}{3}}$$

$$\phi = \tan^{-1}\left(\frac{-11/3}{2}\right) = \tan^{-1}\left(\frac{-11}{6}\right)$$

$$I_0 = \frac{157}{\sqrt{157/3}}$$

$$I_0 = 3\sqrt{157} \approx 37.496A$$

$$I_p(t) = 3\sqrt{56} \sin(3t - 0)$$

$$I_p(t) = 37.5 \sin(3t + 1.67)$$

exact.

$$\left[I_p(t) = 3\sqrt{57} \sin\left(3t + \tan^{-1}\left(\frac{11}{6}\right)\right) \right] \text{Ans.}$$